

Deva Anand

(413) 315-1715 | devaanand@umass.edu | devaanand.com | linkedin.com/in/devaaa | github.com/Deva-1903

Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, TypeScript, JavaScript, Java, C++, SQL, Bash

LLM & Agents

LLM agents, tool use, OpenAI / Anthropic / Groq APIs, provider routing / fallback, prompt design, output verification, evaluation, Ollama, MCP

RAG & Retrieval

RAG over vector stores, dense retrieval, sentence-transformers, FAISS, RRF, cross-encoder reranking, MMR, cell-level provenance, web scraping

Machine Learning

PyTorch, JAX, HuggingFace Transformers, NumPy, scikit-learn, transfer learning, normalizing flows, diffusion, autodiff

Backend & Cloud

FastAPI, Node.js, Express, REST APIs, async services, object-oriented design, Docker, AWS, GCP, Azure, Git, GitHub Actions

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Architected a full-stack **ML training-data annotation platform** (React, Node.js, MongoDB) used daily by the AI team to label data for production models, replacing spreadsheet workflows with secure role-based REST APIs; cut manual annotation work by **100+ hours/month** and improved labeling throughput by **60%**, accelerating iteration on downstream models.
- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions** as a deployed production system; owned reliability and remediated **20+ VAPT findings**.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Integrated **LLMs** into a chat product to generate context-aware replies, iterating on prompt design and response-quality heuristics, while designing **20+ REST APIs** in Node.js + Fastify over PostgreSQL used in production by the frontend team.
- Migrated the core codebase from **JavaScript to TypeScript** across multiple repositories, improving type safety and long-term maintainability through targeted refactoring.

Projects

AgenticSearch - Provenance-First Entity Discovery

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Built a multi-stage **agentic discovery service** that converts open-ended research queries into structured entity tables via typed retrieval planning, page rerank, deterministic extraction with **LLM** fallback, output verification, and **cell-level provenance** for every result, deployed as a live demo on **Google Cloud Run**.
- Hardened the service with **150+ automated tests**, provider routing and fallback across **OpenAI and Groq** APIs, pipeline instrumentation, and a reviewer-facing trust UI for inspecting extraction quality.

KG2RAG-Enhanced - Multi-Hop RAG over a Knowledge Base

github.com/Deva-1903/KG2RAG-685-NLP

Python, Ollama + LLaMA-3, sentence-transformers, llama-index, networkx

- Built the **multi-view retrieval** component of a team KG-guided **RAG** pipeline: query-side decomposition into single-hop sub-questions, per-view **dense retrieval**, and fusion via Reciprocal Rank Fusion, cross-encoder reranking, and MMR, lifting supporting-fact recall **+2.68%** (54.53 to 57.21) over the KG²RAG baseline on a **4,905-question** evaluation.
- Constructed dense-retrieval indices with **sentence-transformer embeddings** over a HotpotQA knowledge base and benchmarked the pipeline across 4,905-question and controlled N=1,000 runs with token and latency logging.

cf-ai-research-scout - Edge-Native AI Research Assistant

github.com/Deva-1903/cf_ai_research_scout

TypeScript, Cloudflare Workers AI, Durable Objects, WebSockets, D1

- Built an edge-native **AI research assistant**: agent fetches web sources, generates multi-step research digests, and streams responses to the client over **WebSocket** in real time, accelerating multi-source research gathering.
- Designed durable workflow boundaries so multi-step research tasks survive process restarts and tool failures, with conversation state persisted server-side without an external database.

Publication & Extracurriculars

- Publication:** “Alzheimer’s Disease Classification using Transfer Learning,” **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023**; contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.